

Utkarsh Shaning

LinkedIn | GitHub | Portfolio | LeetCode

Email : utkarsh1605shaning@gmail.com

Mobile : +91 9201374403

EDUCATION

- **Jaypee University of Information Technology** Solan, Himachal Pradesh
Bachelor of Technology in Computer Science; CGPA: 7.79 2024 – 2028
 - Relevant Coursework: Database Management Systems (DBMS), Object-Oriented Programming (OOPS), Data Structures and Algorithms

EXPERIENCE

- **Research Intern** June 2026 – July 2026
National Institute of Technology Kurukshetra Kurukshetra, Haryana
 - Conducted a systematic literature review on Machine Learning and Deep Learning techniques for aphid and whitefly detection in potato crops under the guidance of Dr. Amandeep Kaur.
 - Collected, filtered, and analyzed research papers from IEEE, ACM Digital Library, Scopus, and Google Scholar to identify state-of-the-art approaches.
 - Prepared structured research findings to support future development of intelligent pest detection systems.

PROJECTS

- **Distributed Task Queue System**
Tech Stack: Java, Spring Boot, RabbitMQ, Docker, AWS EC2 GitHub | Live (guest/guest)
 - Built a high-throughput, distributed message queue for handling asynchronous background tasks efficiently across multiple microservices.
 - Implemented fair-dispatch load balancing across multiple worker nodes using RabbitMQ prefetch and manual ACK, ensuring zero message loss under heavy traffic.
 - Engineered a Dead Letter Queue with exponential backoff retry mechanism for fault tolerance.
- **Scalable Microservices-Based Salon Booking Platform**
Tech Stack: Spring Boot, RabbitMQ, React.js, MongoDB, API Gateway, Stripe, Keycloak, Docker GitHub
 - Built RESTful Spring Boot microservices using layered architecture with Eureka and API Gateway.
 - Developed event-driven workflows using RabbitMQ and integrated Stripe webhooks with MongoDB.
 - Secured distributed services using Keycloak, JWT authentication, and RBAC.
 - Implemented OpenFeign for inter-service communication and containerized the platform using Docker Compose.
- **askPDF**
Tech Stack: FastAPI, LangChain, ChromaDB, Python, Streamlit GitHub | Live
 - Developed RAG-based architecture for contextual question answering over unstructured PDF documents.
 - Implemented document chunking, embedding generation, and semantic similarity search using LangChain and ChromaDB.
 - Exposed inference through FastAPI REST endpoints and built UI for document ingestion and conversational querying.

TECHNICAL SKILLS

- **Languages:** Java, Python
- Frameworks:** Spring Boot, Spring MVC, FastAPI, LangChain
- Databases:** MongoDB, MySQL
- Messaging & Distributed Systems:** RabbitMQ
- Security:** Keycloak, JWT, RBAC, Spring Security
- Dev Tools:** Docker, Postman, Swagger/OpenAPI, Git, Streamlit

HONORS & AWARDS

- **3rd Place** – Cyber AI Hackathon 2025, *University of Derby, UK*, among 100+ teams.